DATASCI 207 • Volatility Dataset — Data Wrangling + EDA

Notebook: `data_wrangler_020426.ipynb`
Last updated: 2026-02-04 21:39 UTC

Goal

Create a **model-ready ETF panel** from the provided volatility dataset exports (CSV/XLSX), with:

- One row per **ETF** per **trading date**
- Clean unique key: **(ticker, date)**
- Macro/risk series transformed from long → wide and **left-joined by date**
- Quality checks to ensure no leakage / no duplicate keys / sensible missingness
- Clean outputs written to `data/clean/`

Inputs (uploaded in this session)

- CSV: `3c3b4b03-7e47-4929-9551-4d6a5edd33d7.csv`
- XLSX: `8ce8cde8-6608-4cb5-a945-638033f95789.xlsx`
- Dataset description PDF: `be3f053b-ba3d-458d-8e6c-2c8b27be8943.pdf`

Dataset facts (from dataset description)

- ~116,525 rows × 18 columns
- 36 ETFs + 6 macro/risk series tickers (42 total)
- Date range: 2014-12-31 to 2025-12-31 (trading days)
- Macro tickers: `US_3M_TB_YLD`, `US_10Y_BOND_YLD`, `US_10Y_3M_YLD_SPREAD`, `NYGOLDS`, `OIL_WTI_S`, `VIX`

---

1. Setup

```
# Purpose:
# - Import common libraries.
# - Set display options.
#
# Algorithm:
# 1) Import numpy/pandas/matplotlib.
# 2) Set pandas display options for readability.

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

pd.set_option("display.max_columns", 200)
pd.set_option("display.width", 140)
```

```
# Mount My Drive:
from google.colab import drive

drive.mount("/content/drive")
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

2. CONFIG (edit here if needed)

```
# Purpose:
# - Centralize all settings in one place.

DATA_PATH_CSV = r"/content/drive/MyDrive/Colab Notebooks/volatility_dataset_013026.csv"
DATA_PATH_XLSX = r"/content/drive/MyDrive/Colab Notebooks/volatility_dataset_013026.xlsx"

# Choose which input to use (CSV recommended for modeling pipelines)
```

```
USE_INPUT = "csv"   # "csv" or "xlsx"

# Key columns (natural primary key)
KEY_COLS = ["ticker", "date"]
DATE_COL = "date"
TARGET_COL = "forward_vol_5d_annual_decimel_calculated"

# Macro/risk series tickers (stored as rows in the same long table; values live in adj_close)
MACRO_TICKERS = [
    "US_3M_TB_YLD",
    "US_10Y_BOND_YLD",
    "US_10Y_3M_YLD_SPREAD",
    "NYGOLDS",
    "OIL_WTI_S",
    "VIX",
]

# Output location
OUTPUT_DIR = "/content/drive/MyDrive/Colab Notebooks/data/clean"
OUTPUT_STEM = "vol_panel_model_ready"

# EDA sampling
EDA_SAMPLE_TICKER = "SPY"
```

3. Load raw data

Gemini

```
# Purpose:
# - Load the dataset into df_raw.
#
# Algorithm:
# 1) Read CSV or XLSX based on USE_INPUT.
# 2) Print shape and preview.

import os
os.makedirs(OUTPUT_DIR, exist_ok=True)

if USE_INPUT.lower() == "csv":
    df_raw = pd.read_csv(DATA_PATH_CSV)
elif USE_INPUT.lower() == "xlsx":
    df_raw = pd.read_excel(DATA_PATH_XLSX)
else:
    raise ValueError("USE_INPUT must be 'csv' or 'xlsx'")

print("Loaded:", USE_INPUT)
print("Shape:", df_raw.shape)
df_raw.head()
```

```
Loaded: csv
Shape: (116525, 18)
/tmp/ipython-input-1200865836.py:12: DtypeWarning: Columns (14,15,16) have mixed types. Specify dtype option on import or set lo
  df_raw = pd.read_csv(DATA_PATH_CSV)
```

	date	ticker	adj_close	has_price	volume_k	volume	trailing_vol_daily_pct_5d_factset	trailing_vol_daily_pct_20d_f
0	12/31/2025	SPY	681.92	1.0	74144.80	74144800.0	0.7149	
1	12/30/2025	SPY	687.01	1.0	47160.67	47160670.0	0.5891	
2	12/29/2025	SPY	687.85	1.0	62559.50	62559500.0	0.5340	
3	12/26/2025	SPY	690.31	1.0	41613.35	41613350.0	0.4783	
4	12/24/2025	SPY	690.38	1.0	39445.56	39445560.0	0.4640	

4. Standardize column names

```
# Purpose:
# - Ensure column names are Python-friendly and consistent.
#
# Algorithm:
# 1) Lowercase
# 2) Replace punctuation/spaces with underscores
```

```
# 3) Collapse multiple underscores

import re

def standardize_column_name(name: str) -> str:
    name = str(name).strip().lower()
    name = re.sub(r"[^0-9a-zA-Z]+", "_", name)
    name = re.sub(r"_+", "_", name)
    return name.strip("_")

df = df_raw.copy()
df.columns = [standardize_column_name(c) for c in df.columns]

print("Columns:", df.columns.to_list())
```

```
Columns: ['date', 'ticker', 'adj_close', 'has_price', 'volume_k', 'volume', 'trailing_vol_daily_pct_5d_factset', 'trailing_vol_d
```

5. Parse dates (mixed formats safe)

```
# Purpose:
# - Parse DATE_COL to true datetime.
# - Handle mixed formats like mm/dd/YYYY and mm/dd/YY.
#
# Algorithm:
# 1) Attempt parse with 4-digit year.
# 2) Fill remaining NaT with 2-digit year parse.
# 3) Report failures and date range.

dt1 = pd.to_datetime(df[DATE_COL], format="%m/%d/%Y", errors="coerce")
dt2 = pd.to_datetime(df[DATE_COL], format="%m/%d/%y", errors="coerce")
df[DATE_COL] = dt1.fillna(dt2)

failed = df[DATE_COL].isna().sum()
print("Failed date parses:", failed)
print("Date range:", df[DATE_COL].min(), "to", df[DATE_COL].max())

# Optional: drop rows with unparseable dates
if failed > 0:
    df = df.dropna(subset=[DATE_COL]).copy()
    print("After dropping failed dates, shape:", df.shape)
```

```
Failed date parses: 0
Date range: 2014-12-31 00:00:00 to 2025-12-31 00:00:00
```

6. Quick QC: keys, duplicates, missingness

```
# Purpose:
# - Sanity checks before transforming macro series.
#
# Algorithm:
# 1) Check duplicates on (ticker, date)
# 2) Missingness summary
# 3) Basic dtypes

dup_count = df.duplicated(subset=KEY_COLS).sum()
print("Duplicate (ticker, date) rows:", dup_count)

missing = df.isna().mean().sort_values(ascending=False)
print("\nMissingness (top 15):")
display((missing.head(15) * 100).to_frame("missing_percent"))

print("\nDtypes:")
display(df.dtypes)
```

```
Duplicate (ticker, date) rows: 0

Missingness (top 15):
```

	missing_percent
market	97.625402
sector	69.131088
industry	47.758850
trailing_vol_annual_decimel_20d_calculated	17.781592
forward_vol_5d_annual_decimel_calculated	17.318172
trailing_vol_annual_decimel_5d_calculated	17.318172
daily_log_return	17.194593
trailing_vol_daily_pct_20d_factset	17.194593
trailing_vol_annual_decimel_20d_factset	17.194593
volume	17.163699
trailing_vol_annual_decimel_5d_factset	17.163699
trailing_vol_daily_pct_5d_factset	17.163699
volume_k	17.163699
has_price	14.515340
group_type	14.515340

```
Dtypes:
```

	0
date	datetime64[ns]
ticker	object
adj_close	float64
has_price	float64
volume_k	float64
volume	float64
trailing_vol_daily_pct_5d_factset	float64
trailing_vol_daily_pct_20d_factset	float64
trailing_vol_annual_decimel_5d_factset	float64
trailing_vol_annual_decimel_20d_factset	float64
trailing_vol_annual_decimel_5d_calculated	float64
trailing_vol_annual_decimel_20d_calculated	float64
forward_vol_5d_annual_decimel_calculated	float64
group_type	object
market	object
sector	object
industry	object
daily_log_return	float64

7. Split ETFs vs Macro series

```
# Purpose:
# - Separate ETF panel rows from macro/risk rows.
#
# Definitions:
# - Macro rows: ticker in MACRO_TICKERS. Value stored in adj_close; other columns often null.
#
# Output:
# - df_etf_long: ETF-only long panel
# - df_macro_long: macro-only long table

df["ticker"] = df["ticker"].astype(str)

df_macro_long = df[df["ticker"].isin(MACRO_TICKERS)].copy()
```

```
df_etf_long = df[~df["ticker"].isin(MACRO_TICKERS)].copy()

print("ETF rows:", df_etf_long.shape)
print("Macro rows:", df_macro_long.shape)

print("\nMacro tickers present:", sorted(df_macro_long["ticker"].unique().tolist()))
```

```
ETF rows: (102482, 18)
Macro rows: (14043, 18)

Macro tickers present: ['NYGOLDS', 'OIL_WTI_S', 'US_10Y_BOND_YLD', 'US_3M_TB_YLD', 'VIX']
```

8. Macro: long → wide and join onto ETFs by date

```
# Purpose:
# - Transform macro series from rows to columns and left-join onto ETF panel by date.
#
# Algorithm:
# 1) Create wide macro table: index = date, columns = macro ticker, values = adj_close
# 2) Sort by date and forward-fill missing macro values
# 3) Left-join onto ETF panel by date

# Step 1: pivot
macro_wide = (
    df_macro_long
    .pivot_table(index=DATE_COL, columns="ticker", values="adj_close", aggfunc="first")
    .sort_index()
)

# Step 2: forward-fill (common for rates/VIX if a macro value missing on a trading date)
macro_wide = macro_wide.ffill()

print("macro_wide shape:", macro_wide.shape)
display(macro_wide.tail(5))

# Step 3: join
df_etf_long = df_etf_long.sort_values(["ticker", DATE_COL]).copy()
df_panel = df_etf_long.merge(macro_wide, how="left", left_on=DATE_COL, right_index=True)

print("df_panel shape:", df_panel.shape)
df_panel.head()
```

```
macro_wide shape: (2871, 5)
```

ticker	NYGOLDS	OIL_WTI_S	US_10Y_BOND_YLD	US_3M_TB_YLD	VIX
date					
2025-12-25	4479.42	58.72	4.13	3.64	13.47
2025-12-26	4533.21	56.60	4.13	3.63	13.60
2025-12-29	4332.56	57.89	4.11	3.62	14.20
2025-12-30	4339.40	57.79	4.13	3.60	14.33
2025-12-31	4322.61	57.26	4.17	3.63	14.95

```
df_panel shape: (102482, 23)
```

	date	ticker	adj_close	has_price	volume_k	volume	trailing_vol_daily_pct_5d_factset	trailing_vol_daily_pct_20d_facts
63639	2014-12-31	BEDZ	NaN	0.0	NaN	NaN	NaN	N
63638	2015-01-02	BEDZ	NaN	0.0	NaN	NaN	NaN	N
63637	2015-01-05	BEDZ	NaN	0.0	NaN	NaN	NaN	N
63636	2015-01-06	BEDZ	NaN	0.0	NaN	NaN	NaN	N
63635	2015-01-07	BEDZ	NaN	0.0	NaN	NaN	NaN	N

9. Enforce modeling filters (has_price, target not-null)

```
# Purpose:
# - Create a clean modeling subset.
#
# Recommended rules (dataset description):
# - Use only rows where has_price == 1
# - Ensure TARGET_COL not null for training rows
#
# Notes:
# - The last 5 trading days for each ticker will have null forward 5d volatility by construction.

df_panel["has_price"] = pd.to_numeric(df_panel.get("has_price", 1), errors="coerce")

df_model = df_panel.copy()

# Filter has_price if column exists
if "has_price" in df_model.columns:
    df_model = df_model[df_model["has_price"] == 1].copy()

print("After has_price filter:", df_model.shape)

# Report target missingness (do not drop by default; tail rows per ticker will be missing)
if TARGET_COL in df_model.columns:
    tgt_missing = df_model[TARGET_COL].isna().mean()
    print("Target missing rate:", round(float(tgt_missing), 4))
else:
    print("WARNING: TARGET_COL not found in columns.")
```

```
After has_price filter: (96525, 23)
Target missing rate: 0.0019
```

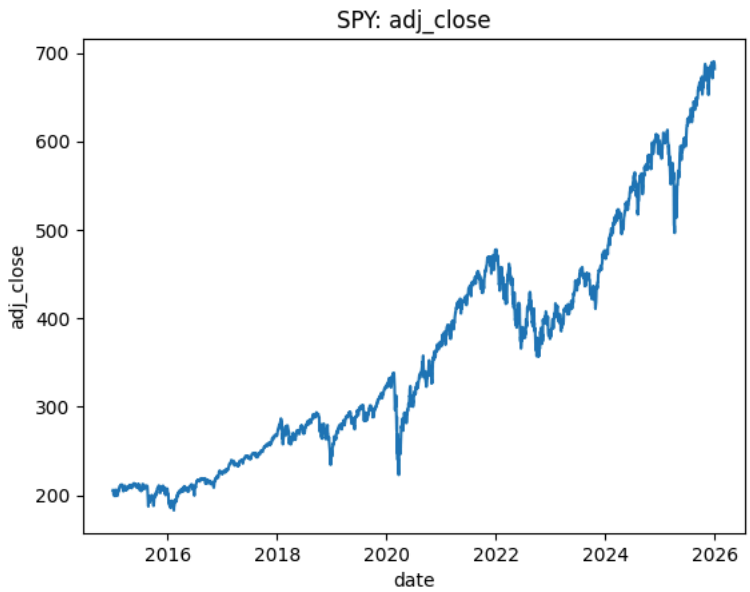
10. Key QC checks (no duplicates, monotone dates within ticker)

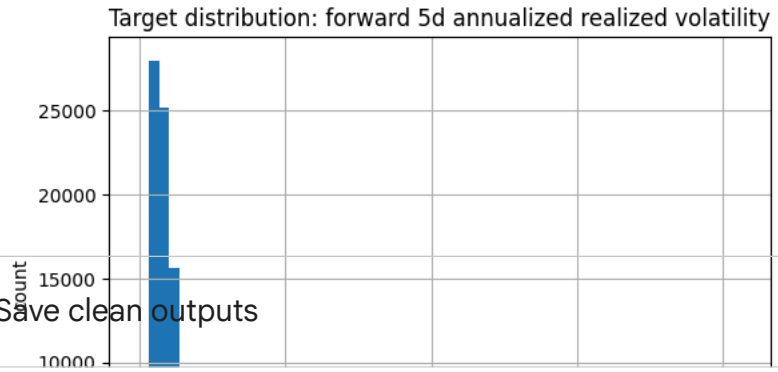
```
# Purpose:
# - Verify panel integrity.
#
# Algorithm:
# 1) No duplicate (ticker,date)
# 2) Dates increasing within each ticker
# 3) Quick spot-check of daily_log_return vs adj_close

dup_count = df_model.duplicated(subset=KEY_COLS).sum()
print("Duplicate keys in df_model:", dup_count)

# monotone increasing check
bad = 0
for tkr, g in df_model.groupby("ticker"):
    d = g[DATE_COL]
    if not d.is_monotonic_increasing:
        bad += 1
        if bad <= 5:
            print("Non-monotone dates for ticker:", tkr)

print("Tickers with non-monotone dates:", bad)

# Spot-check: daily_log_return sanity for sample ticker
if "adj_close" in df_model.columns and "daily_log_return" in df_model.columns:
    sample = df_model[df_model["ticker"] == EDA_SAMPLE_TICKER].sort_values(DATE_COL).copy()
    # recompute returns from adj_close
    recomputed = np.log(sample["adj_close"]).diff()
    diff_abs = (recomputed - sample["daily_log_return"]).abs()
    print("Return check (median abs diff):", float(diff_abs.median(skipna=True)))
    print("Return check (max abs diff):", float(diff_abs.max(skipna=True)))
else:
    print("Skipping return spot-check (missing columns).")
```

```
Duplicate keys in df_model: 0
Tickers with non-monotone dates: 0
Return check (median abs diff): 2.5422055346252526e-10
Return check (max abs diff): 2.7679655563378393e-05
```

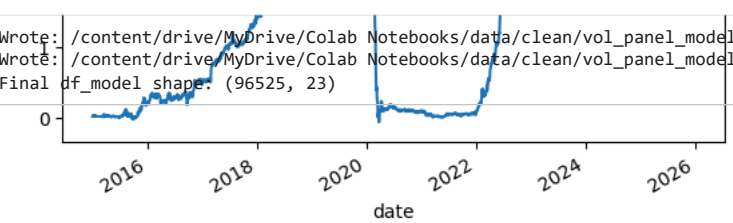
11. Lightweight EDA (sanity plots)

```
# Purpose:
# - Quick EDA plots for intuition and sanity checks.
#
# Notes:
# - Keep this lightweight; heavy EDA belongs in a separate notebook later if needed.

sample = df_model[df_model["ticker"] == EDA_SAMPLE_TICKER].sort_values(DATE_COL).copy()

# Price over time
if "adj_close" in sample.columns:
    plt.figure()
    plt.plot(sample[DATE_COL], sample["adj_close"])
    plt.title(f"{EDA_SAMPLE_TICKER}: adj_close")
    plt.xlabel("date")
    plt.ylabel("adj_close")
    plt.show()

# Target distribution
if TARGET_COL in df_model.columns:
    plt.figure()
    df_model[TARGET_COL].dropna().hist(bins=60)
    plt.title("Target distribution: forward 5d annualized realized volatility")
    plt.xlabel(TARGET_COL)
    plt.ylabel("count")
    plt.show()

# Macro series quick look
macro_cols = [c for c in MACRO_TICKERS if c in df_model.columns]
if macro_cols:
    plt.figure()
    df_model.sort_values(DATE_COL).groupby(DATE_COL)[macro_cols[0]].mean().plot()
    plt.title(f"Macro series example: {macro_cols[0]}")
    plt.xlabel("date")
    plt.ylabel("value")
    plt.show()
```





12. Save clean outputs

```
# Purpose:
# - Save the model-ready panel for downstream modeling.
#
# Outputs:
# - Parquet (preferred)
# - CSV (optional)

out_parquet = os.path.join(OUTPUT_DIR, f"{OUTPUT_STEM}.parquet")
out_csv = os.path.join(OUTPUT_DIR, f"{OUTPUT_STEM}.csv")

# Write Parquet if available
try:
    df_model.to_parquet(out_parquet, index=False)
    print("Wrote:", out_parquet)
except Exception as e:
    print("Parquet write failed (install pyarrow). Error:", e)

df_model.to_csv(out_csv, index=False)
print("Wrote:", out_csv)

print("Final df_model shape:", df_model.shape)
```

```
Wrote: /content/drive/MyDrive/Colab Notebooks/data/clean/vol_panel_model_ready.parquet
Wrote: /content/drive/MyDrive/Colab Notebooks/data/clean/vol_panel_model_ready.csv
Final df_model shape: (96525, 23)
```